as the angle formed by BC tand BE. Therefore $\angle DAF = \angle CBE$. Thus ΔADF and ΔBCE are congruent.

That is an interesting observation; can we infer anything more from this ? Yes, we know that congruent triangles have the *same area*. This makes us think about the areas of the parallelograms $ABCD$ and $ABEF$.

$$\begin{aligned}\text{Area of } ABCD &= \text{area of quadrilateral } ABED + \text{area of } \Delta BCE \\ &= \text{area of quadrilateral } ABED + \text{area of } \Delta ADF \\ &= \text{area of } ABEF\end{aligned}$$

Thus we have proved another interesting theorem:

Theorem 5:

Parallelograms on the same base and between the same parallels are equal in area.

In this process, we have also proved other interesting statements. These are called *Corollaries*, which do not need separate detailed proofs.

Corollary 1: Triangles on the same base and between the same parallels are equal in area.

Corollary 2: A rectangle and a parallelogram on the same base and between the same parallels are equal in area.

These statements that we called Theorems and Corollaries, hold for all parallelograms, however large or small, with whatever be the lengths of sides and angles at vertices.

Example 4.1

In a parallelogram $ABCD$, the bisectors of the consecutive angles $\angle A$ and $\angle B$ insersect at P. Show that $\angle APB = 90°$

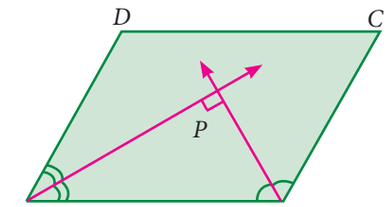


Fig. 4.34

Solution

$ABCD$ is a parallelogram AP and BP are bisectors of consecutive angles $\angle A$ and $\angle B$.

Since the consecutive angles of a parallelogram are supplementary

$$\angle A + \angle B = 180°$$

$$\frac{1}{2}\angle A + \frac{1}{2}\angle B = \frac{180^{o}}{2}$$

$$\Longrightarrow \angle PAB + \angle PBA = 90°$$

In ΔAPB,

$$\angle PAB + \angle APB + \angle PBA = 180° \text{ (angle sum property of triangle)}$$

$$\angle APB = 180° - [\angle PAB + \angle PBA]$$

$$= 180° - 90° = 90°$$

Hence Proved.

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